

Segmentation Fault in GrammarMatcherBase::ExpandEquivalentStackElements #250

Closed

Saibo-creator opened on Mar 18, 2025 · edited by Saibo-creator

Edits · ···

I'm encountering a **segmentation fault (SIGSEGV)** when using the grammar matcher module. The error occurs inside: `xgrammar::GrammarMatcherBase::ExpandEquivalentStackElements` at `grammar_matcher_base.cc:179`

Reproduce

```
import xgrammar as xgr
from json import dumps
from transformers import AutoTokenizer, AutoConfig

device = "cuda"
model_name = "meta-llama/Llama-3.2-1B-Instruct"
tokenizer = AutoTokenizer.from_pretrained(model_name)
config = AutoConfig.from_pretrained(model_name)

json_schema= {"$schema": "http://json-schema.org/draft-04/schema#", "type": "object", "required": ["text"], "properties": {"text": {"type": "string"}}}

tokenizer_info = xgr.TokenizerInfo.from_huggingface(tokenizer, vocab_size=config.vocab_size)
grammar_compiler = xgr.GrammarCompiler(tokenizer_info, max_threads=1)
compiled_grammar = grammar_compiler.compile_json_schema(schema=dumps(json_schema))

print(compiled_grammar)
```

GDB Backtrace

```
Thread 1 "python" received signal SIGSEGV, Segmentation fault.
0x00007ffffa2886da2 in xgrammar::GrammarMatcherBase::ExpandEquivalentStackElements
(this=this@entry=0x7fffffbfbf30, cur_stack_element=...,
new_stack_tops=new_stack_tops@entry=0x7fffffbfd8,
cur_stack_element_id=cur_stack_element_id@entry=-1,
consider_parent=consider_parent@entry=false) at /project/cpp/grammar_matcher_base.cc:179
warning: 179 /project/cpp/grammar_matcher_base.cc: No such file or directory
#0  0x00007ffffa2886da2 in xgrammar::GrammarMatcherBase::ExpandEquivalentStackElements at
grammar_matcher_base.cc:179
#1  0x00007ffffa2886fb0 in xgrammar::GrammarMatcherBase::ExpandEquivalentStackElements at
grammar_matcher_base.cc:205
#2  0x00007ffffa2887041 in xgrammar::GrammarMatcherBase::ExpandEquivalentStackElements at
grammar_matcher_base.cc:247
#3  0x00007ffffa2886e77 in xgrammar::GrammarMatcherBase::ExpandEquivalentStackElements at
```

```
grammar_matcher_base.cc:271
...
```

Environemnt

- Python | 3.12
- xgrammar | 0.1.16

P.S. The grammar is taken from JSONSchemaBench: https://github.com/guidance-ai/jsonschemabench/blob/main/data/Github_medium/o65372.json

Ubospica on Mar 19, 2025

Collaborator ...

Thanks for the report! We will take a look

benchisleett on Apr 29, 2025

Contributor ...

I am also seeing this error, any ideas?

```
Caught signal 11 (Segmentation fault: address not mapped to object at address
0x7fff5063ffec)
==== backtrace (tid: 79508) ====
 0 0x00000000000042520 __sigaction() ???:0
 1 0x0000000000005869c xgrammar::GrammarMatcherBase::ExpandEquivalentStackElements()
/project/cpp/grammar_matcher_base.cc:171
 2 0x00000000000058822 xgrammar::GrammarMatcherBase::ExpandEquivalentStackElements()
/project/cpp/grammar_matcher_base.cc:205
 3 0x000000000000588b1 xgrammar::GrammarMatcherBase::ExpandEquivalentStackElements()
/project/cpp/grammar_matcher_base.cc:247
 4 0x000000000000588b1 xgrammar::Grammar::operator->()
/project/include/xgrammar/grammar.h:156
 5 0x000000000000588b1 xgrammar::GrammarMatcherBase::ExpandEquivalentStackElements()
/project/cpp/grammar_matcher_base.cc:269
 6 0x0000000000005877a xgrammar::GrammarMatcherBase::ExpandEquivalentStackElements()
/project/cpp/grammar_matcher_base.cc:236
 7 0x0000000000005877a xgrammar::GrammarMatcherBase::ExpandEquivalentStackElements()
/project/cpp/grammar_matcher_base.cc:236
 8 0x0000000000005877a xgrammar::GrammarMatcherBase::ExpandEquivalentStackElements()
/project/cpp/grammar_matcher_base.cc:236
 9 0x0000000000005877a xgrammar::GrammarMatcherBase::ExpandEquivalentStackElements()
/project/cpp/grammar_matcher_base.cc:236
10 0x0000000000005877a xgrammar::GrammarMatcherBase::ExpandEquivalentStackElements()
/project/cpp/grammar_matcher_base.cc:236
11 0x0000000000005877a xgrammar::GrammarMatcherBase::ExpandEquivalentStackElements()
/project/cpp/grammar_matcher_base.cc:236
12 0x0000000000005877a xgrammar::GrammarMatcherBase::ExpandEquivalentStackElements()
/project/cpp/grammar_matcher_base.cc:236
13 0x0000000000005877a xgrammar::GrammarMatcherBase::ExpandEquivalentStackElements()
/project/cpp/grammar_matcher_base.cc:236
14 0x0000000000005877a xgrammar::GrammarMatcherBase::ExpandEquivalentStackElements()
/project/cpp/grammar_matcher_base.cc:236
15 0x0000000000005877a xgrammar::GrammarMatcherBase::ExpandEquivalentStackElements()
/project/cpp/grammar_matcher_base.cc:236
16 0x0000000000005877a xgrammar::GrammarMatcherBase::ExpandEquivalentStackElements()
```



[illegible]

```
49 0x0000000000005877a xgrammar::GrammarMatcherBase::ExpandEquivalentStackElements()  
/project/cpp/grammar_matcher_base.cc:236  
50 0x0000000000005877a xgrammar::GrammarMatcherBase::ExpandEquivalentStackElements()  
/project/cpp/grammar_matcher_base.cc:236  
51 0x0000000000005877a xgrammar::GrammarMatcherBase::ExpandEquivalentStackElements()  
/project/cpp/grammar_matcher_base.cc:236  
52 0x0000000000005877a xgrammar::GrammarMatcherBase::ExpandEquivalentStackElements()  
/project/cpp/grammar_matcher_base.cc:236  
53 0x0000000000005877a xgrammar::GrammarMatcherBase::ExpandEquivalentStackElements()  
/project/cpp/grammar_matcher_base.cc:236  
54 0x0000000000005877a xgrammar::GrammarMatcherBase::ExpandEquivalentStackElements()  
/project/cpp/grammar_matcher_base.cc:236  
55 0x0000000000005877a xgrammar::GrammarMatcherBase::ExpandEquivalentStackElements()  
/project/cpp/grammar_matcher_base.cc:236  
56 0x0000000000005877a xgrammar::GrammarMatcherBase::ExpandEquivalentStackElements()  
/project/cpp/grammar_matcher_base.cc:236  
=====
```

Seven-Streams on May 13, 2025

Collaborator ...

It seems like there is a infinite recursion in the grammar:

```
root_prop_4_additional_prop_1_8 ::= (" | ([/a-zA-Z0-9_ .\ -]*  
root_prop_4_additional_prop_1_8)) (=(root_prop_4_additional_prop_1_9 "\""))
```



It seems like the issue will be resolved by [#308](#).

3



halaction mentioned this [on Jun 30, 2025](#)



[\[Bug\] Deployment Crash due to Recursive Grammar in Structured Outputs sgl-project/sglang#7639](#)



Ubospica added **bug** [on Jun 30, 2025](#)

Ubospica on Jul 12, 2025

Collaborator ...

This should have been fixed in [#308](#) and v0.1.21. Feel free to reopen it if the problem still occurs.



Ubospica closed this as [completed](#) [on Jul 12, 2025](#)

Sign up for free to join this conversation on GitHub. Already have an account? [Sign in to comment](#)

Assignees

No one assigned

Labels

bug

Type

No type

Projects

No projects

Milestone

No milestone

Relationships

None yet

Development

 Code with agent mode

▼

No branches or pull requests

Participants

